

Forest Edge Effects on Plant Biodiversity

Research question: How does distance from the forest edge affect understory plant cover, biodiversity, and the relative abundance of invasive species in Pacific Spirit Regional Park?

Overview

Investigated forest-edge effects on understory plant communities in Pacific Spirit Regional Park. Field sampling was conducted along a forest transect to document plant species and percent cover, followed by analysis of pooled class data to examine how total plant cover, biodiversity, and invasive-species abundance changed with increasing distance from the forest edge.

Field Methods

- Established a 20 m transect perpendicular to the trail and sampled vegetation along an edge-to-interior gradient.
- Surveyed 11 1 m × 1 m quadrats at 2 m intervals, identifying understory plant species and estimating their percent cover.
- Used consistent percent-cover estimation procedures and plant identification resources to collect comparable vegetation data across quadrats.
- Combined group field observations with pooled class data to investigate broader ecological patterns with distance from the forest edge.

Data Analysis

- Created a species-accumulation curve to assess whether the transect captured most of the plant species present.
- Calculated average plant cover across forest-edge distances using pooled class data.
- Calculated Shannon diversity (H) to evaluate biodiversity based on species richness and relative abundance.
- Calculated the relative abundance of invasive species and compared it with distance from the forest edge.
- Used trendlines, R^2 values, and Pearson correlation coefficients to interpret ecological relationships.

Results

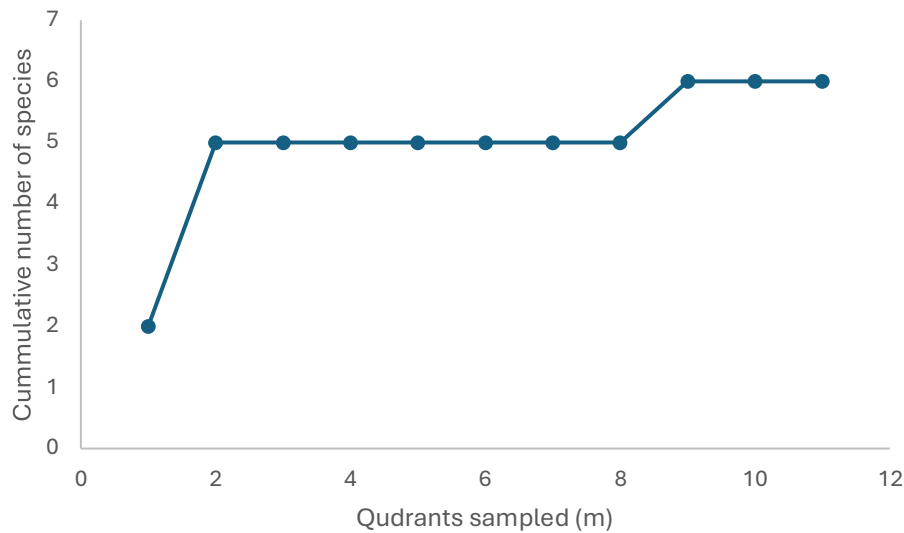


Figure 1. Species-accumulation curve showing the cumulative number of unique understory plant species detected across 11 quadrats along the sampled transect. The curve began to level off as additional quadrats were sampled, suggesting that the transect captured many of the species present in the sampled area.

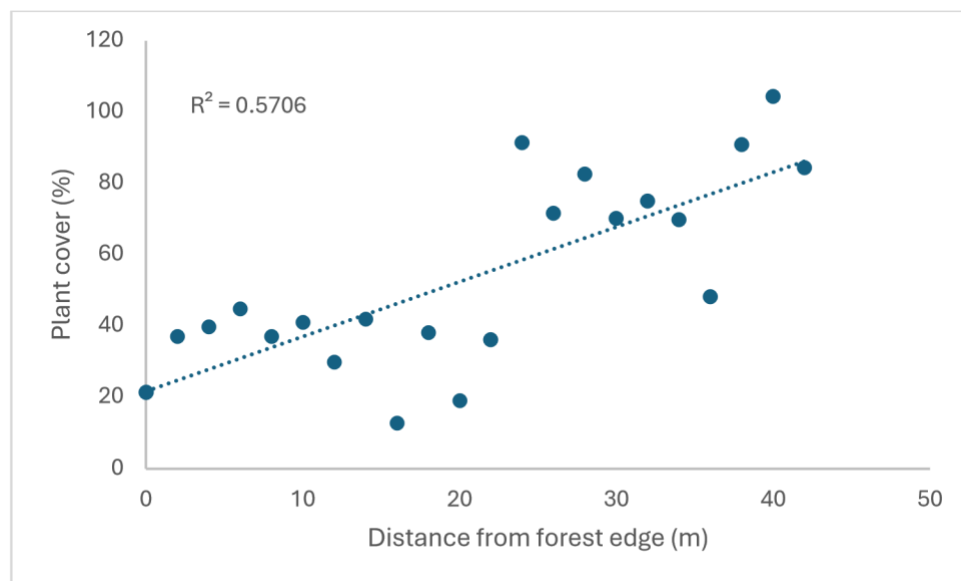


Figure 2. Relationship between total understory plant cover and distance from the forest edge. Plant cover increased with greater distance from the edge ($R^2 = 0.571$, $r = 0.755$), indicating greater vegetation cover toward the forest interior.

Result: Distance from the forest edge explained approximately 57% of the variation in plant cover, with greater plant cover generally observed farther into the forest.

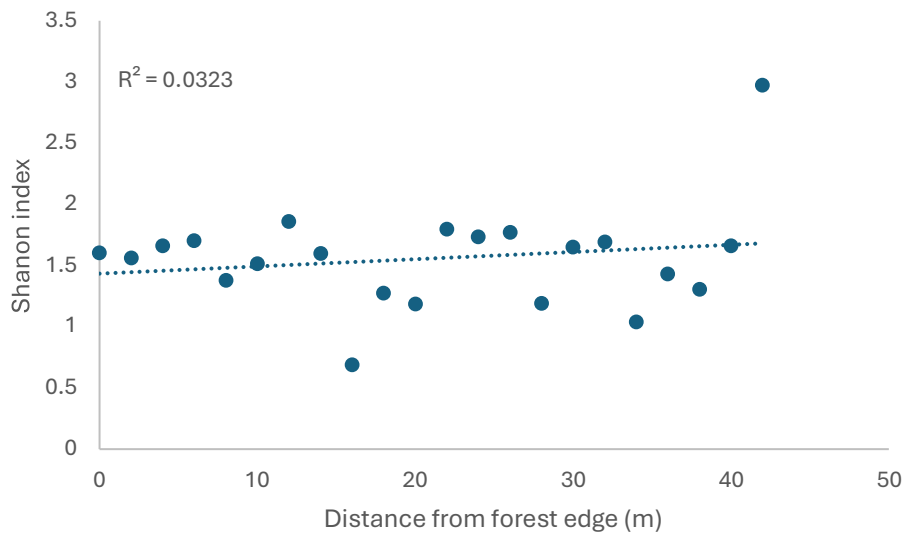


Figure 3. Relationship between Shannon diversity and distance from the forest edge. Shannon diversity showed little relationship with forest-edge distance ($R^2 = 0.032$, $r = 0.18$), with $H = 2.97$ at 42 m.

Result: Biodiversity remained relatively similar across the forest-edge gradient, with distance explaining little of the observed variation in Shannon diversity.

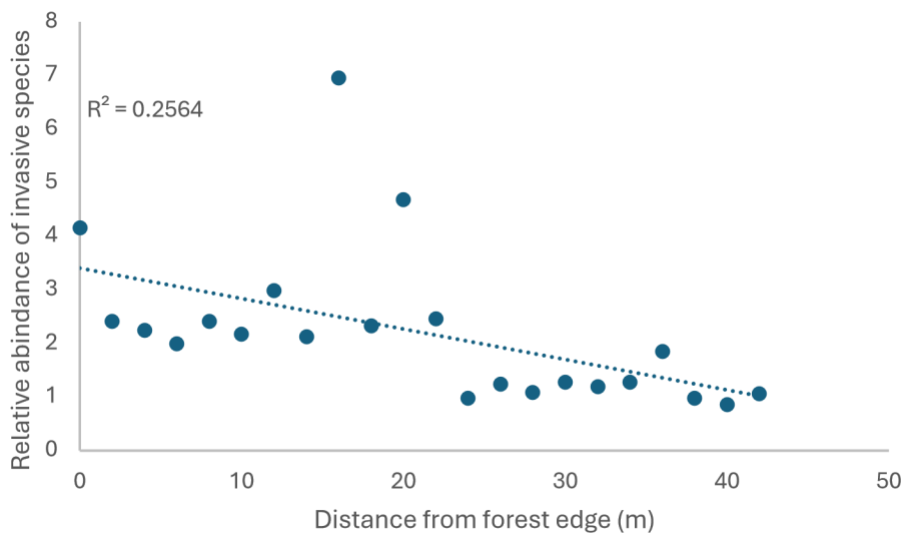


Figure 4. Relationship between the relative abundance of invasive plant species and distance from the forest edge. Invasive-species abundance declined with greater distance from the edge ($R^2 = 0.256$, $r = -0.51$).

Result: Invasive species were generally more abundant near the forest edge and declined toward the forest interior.

Interpretation

Total plant cover increased with distance from the forest edge, while Shannon diversity showed little change. These patterns are not necessarily contradictory because greater vegetation cover does not automatically indicate greater biodiversity. A location may contain more total plant cover while maintaining a similar number and distribution of species.

In contrast, the relative abundance of invasive species declined with distance from the forest edge. This pattern was consistent with the expectation that forest-edge disturbance can influence species composition even when overall community diversity remains relatively stable.

Conclusion

Distance from the forest edge was associated with clear changes in understory vegetation at Pacific Spirit Regional Park. Plant cover increased toward the forest interior, while invasive-species relative abundance declined. In contrast, Shannon diversity changed little across the gradient, suggesting that forest-edge effects influenced vegetation abundance and community composition more strongly than overall plant diversity in this dataset.

Limitations

Percent cover was visually estimated and may therefore contain observer variation. The group transect also represented a limited spatial sample, while environmental variables such as light, moisture, trampling, and pollution were not individually tested. Future sampling across additional transects and seasons could help determine whether the observed edge-effect patterns remain consistent over larger spatial and temporal scales.